

5(a)(i)	<i>two from:</i> 1 CAG / repeat, is a triplet or - triplets of bases / codons, are added ; 2 (so) other triplets are unchanged ; 3 (so) no frameshift ;	2
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Question	Answer	Marks
5(a)(ii)	<i>four from:</i> 1 allele is dominant ; 2 so will still be expressed even when, normal / recessive, allele is present ; ora 3 (gene therapy only) used to treat recessive (allele) disorders ; 4 cannot, remove dominant allele / replace an allele ; 5 dominant allele affects tissues in many parts of the body ;	4
5(b)(i)	50% / 1 in 2 / 0.5 / half ; A 100% if candidate implies parent is homozygous dominant	1
5(b)(ii)	<i>two from:</i> <i>advantage (max 1)</i> 1 can choose whether to have children ; 2 can prepare for the future ; 3 (if negative) removes anxiety / AW ; <i>not awarded with mp5</i> <i>disadvantage (max 1)</i> 4 (if positive) no treatment possible ; 5 (if positive) may lead to anxiety / AW ; <i>not awarded with mp3</i> 6 (if positive) social / financial, discrimination ; e.g. life insurance refusal 7 (if positive) may still not develop disease (<i>ref. to below 39 repeats</i>) ;	2
5(c)(i)	removing a cell from an embryo (for testing) ;	1
5(c)(ii)	to amplify the, DNA / gene, (from the embryo cell) ;	1

Question	Answer	Marks
5(c)(iii)	<i>two from:</i> 1 embryos might be destroyed ; 2 wrong for parents to choose / designer embryos ; 3 contrary to, beliefs / values ; 4 less chance of Huntington allele being passed on / decrease in frequency of Huntington allele ; 5 <i>idea that</i> people with the faulty allele who otherwise would not have children can now do so ;	2